

Math 372: Homework 12

Due Wednesday, May 6

Instructions: Do not write your solutions on this sheet—you don't have enough space to show your work. So use separate sheets of paper. For all problems you need to show your work.

Problem 1. Consider the following three data points, of the form (x, y) :

$$(-1, 4), \quad (0, 5), \quad \text{and} \quad (1, 9)$$

- (a) Plot the data points on the xy -plane
- (b) Suppose we have a line $y = C + Dx$. Let $\vec{e} = (e_1, e_2, e_3)$ be the vector whose entries are the vertical distances from the data points to the line. Write out $\vec{e}$.
- (c) Find the values of C and D which minimize the length of $\|\vec{e}\|^2 = e_1^2 + e_2^2 + e_3^2$.
- (d) Sketch the line $y = C + Dx$ using your values of C and D .

Problem 2 (Salk Polio Field Trial (1954)). The results of a 1954 polio vaccine trial in $n = 401,974$ elementary school students gave the following results:

	No Polio	Polio
Vaccinated	200,712	33
Unvaccinated	201,114	115

(1)

We wish to know if there is a relationship between vaccination status and the incidence of paralytic polio. The null hypothesis is:

$$H_0 = \text{vaccination and polio are independent}$$

and the alternative hypothesis is

$$H_a = \text{there is a relationship between vaccination and polio.}$$

Conduct a hypothesis test using the chi-squared statistic to test these hypotheses.

Problem 3. Let S_{200} be the number of heads that turn up in 200 tosses of a fair coin. Estimate

$$\mathbb{P}[S_{200} = 100], \mathbb{P}[S_{200} = 90], \text{ and } \mathbb{P}[S_{200} = 80].$$

Problem 4. Find the probability that among 10,000 random digits the digit 3 appears not more than 931 times.

Problem 5. A random walker starts at 0 on the x-axis and at each time unit moves 1 step to the right or 1 step to the left with probability $1/2$. Estimate the probability that, after 100 steps, the walker is more than 10 steps from the starting position.

Problem 6. An urn contains one red ball and one blue ball. A ball is drawn at random. If it is red, the game is over. If it is blue, it is replaced in the urn along with an extra blue ball. This process continues until the red ball is drawn, however long it takes. Let Y be the number of draws required.

- (a) Argue that $\mathbb{P}[Y = k] = \frac{1}{k(k+1)}$.
- (b) Show that this defines a valid pmf. (Hint: first show that $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, and then consider $\sum_{k=1}^n \frac{1}{k(k+1)}$).
- (c) Show that $\mathbb{E}[Y]$ does not exist.

Problem 7. Consider the following four datapoints:

$$(-1, 2), \quad (0, 0), \quad (1, -3), \quad \text{and} \quad (2, -5)$$

(a) Find the best straight-line fit to the data.

(b) Plot points and the line.

Problem 8 (Normal Distribution). If $Z \sim \mathcal{N}(0, 1)$, then

$$\mathbb{P}[|Z| \leq u] = 2\mathbb{P}[Z \leq u] - 1 \tag{2}$$

for any $u \geq 0$. Draw a picture to illustrate the geometric meaning of Eq. (2).